

Guide to the PAC Dashboards

Part 1: How political money works

Before the charts make sense, it helps to understand who the players are and how money moves between them. This part explains the whole system from scratch. If you already know it, skip to Part 3.

The players

Candidate: a person running for federal office like President, U.S. Senate, or U.S. House of Representatives. Each candidate has an official campaign committee that collects and spends money for them.

Committee: any organization registered with the Federal Election Commission (FEC) that raises or spends political money. Campaigns, PACs, and Super PACs are all 'committees.'

FEC: the Federal Election Commission means the U.S. government agency that collects and publishes all this money data. Every dollar in these dashboards comes from FEC filings.

PAC vs Super PAC — the key distinction

This is the single most important concept. There are two very different kinds of political organizations, and they move money in opposite ways.

A regular PAC (Political Action Committee)

A PAC collects money from many people (for example, the employees of a company, or members of a union or trade group) and then GIVES that money directly to candidates it supports. A company like Torch Technologies has a corporate PAC; its employees chip in, and the PAC writes checks to candidates.

The crucial limit: a PAC can only give a candidate a small, capped amount like a few thousand dollars per election. So PAC contributions are modest in size. They're about building relationships and access, not single-handedly funding a campaign.

A regular PAC GIVES money TO candidates

Money flows: company employees → corporate PAC → candidate. The amounts are small and capped. This is a 'contribution.' Our original three dashboards (industry_rankings, monthly_by_firm, election_comparison) are about this kind of money.

A Super PAC (independent-expenditure-only committee)

A Super PAC is different. It can raise UNLIMITED money from corporations, unions, and wealthy individuals but it is NOT allowed to give that money to candidates. Instead, it spends the money itself, independently, on advertising that supports or attacks candidates. It buys TV ads, digital ads, direct mail, and so on.

Because there's no limit on what a Super PAC can raise and spend, this is where the large dollar figures live such as hundreds of millions of dollars per Super PAC. The catch (legally) is

that a Super PAC must operate 'independently' and cannot coordinate with the candidate it's helping.

A Super PAC SPENDS money ON ads, it does NOT give to candidates

Money flows: rich donors/corporations → Super PAC → advertising vendors (who make and run the ads). The candidate never touches the money. This is an 'independent expenditure.' Our updated file is almost entirely this kind of money.

Two kinds of transactions — this is what the data records

Every row in the data is one transaction, and the FEC tags each with a transaction-type code. Three matter most:

24K — Direct contribution: a PAC gives money directly to a candidate. Small, capped amounts. This is the 'PAC gives to candidate' money.

24E — Independent expenditure FOR: a Super PAC spends money on advertising that SUPPORTS a candidate (a positive ad). The money goes to an ad vendor, not the candidate.

24A — Independent expenditure AGAINST: a Super PAC spends money on advertising that ATTACKS a candidate (a negative/attack ad). Again, the money goes to a vendor.

So '24E for Kamala Harris' means a Super PAC paid for ads boosting Harris. '24A against Kamala Harris' means a Super PAC paid for ads attacking Harris. The same candidate can have huge amounts of both (for example, money spent to help her AND money spent to tear her down).

Primary vs General election

U.S. elections happen in two rounds, and money is tracked separately for each:

Primary election: the first round, held within a single party, where voters choose which candidate will represent that party. (E.g., Trump vs. Haley vs. DeSantis for the Republican nomination.)

General election: the final round in November, where the parties' winners face each other. (E.g., Trump vs. Harris.)

The FEC tags each transaction with a 'PGI' (primary/general indicator): P for primary, G for general. Contribution limits reset between the two rounds, which is why the distinction is tracked.

Where do companies fit in? (Two completely different roles)

This trips people up, so read carefully. Companies appear in this data in TWO unrelated ways:

- As DONORS: a company's corporate PAC gives money to candidates (e.g., Torch Technologies' PAC contributes to a congressman). This is the 'industry rankings' story — classifying donor companies by industry (Defense, Legal, Healthcare...).
- As VENDORS: a Super PAC PAYS a company to produce or run its ads (e.g., paying Google for digital ads, a print shop for mailers, an ad agency for TV spots). These companies aren't donating — they're being hired. This is the 'vendor industries' story.

Same company name, opposite meaning

Meta (Facebook) is a perfect example. Meta does NOT appear as a donor in our data — it didn't give money to candidates. But Meta appears as a VENDOR: Super PACs paid Meta \$10.3M for digital ads. So Meta shows up in the vendor analysis, not the contribution/industry rankings. Always ask: is this company giving money, or getting paid?

Federal contractors and SAM.gov

SAM.gov is the U.S. government's official registry of companies that hold federal contracts. The updated data tries to match the ad vendors against this registry, to flag which vendors are also federal contractors (an interesting question: are firms that get government contracts also profiting from political ad spending?).

Important caution that recurs in the dashboards: this matching is imperfect. A short or common company name can accidentally match the wrong registered company. We found, for example, that 'META' got matched to an unrelated social-services company called 'META LLC.' So the contractor labels need a skeptical eye — a perfect name match is not the same as the right company.

Part 2 — The two datasets behind the dashboards

There are two underlying data files, and they tell different stories. Knowing which file a dashboard uses tells you what its numbers mean.

The original file: `pac_with_firms.parquet` (contributions)

This is corporate-PAC contribution data: money GIVEN to candidates. It came with a hand-built 'industry' label for each donor firm (Defense, Legal, Healthcare, etc.). The total matched money here is small (under \$1M) because direct PAC contributions are capped and modest. The original three dashboards were built on this.

The updated file: `pac_merged.parquet` (independent expenditures)

This is the new file your colleague provided. It is overwhelmingly (93%) independent-expenditure data like Super PAC ad SPENDING, not contributions. It totals about \$4.49 billion. It adds federal-contractor (SAM.gov) matching and near-complete candidate information, but it has NO hand-built donor-industry column. The five newer dashboards use this file.

The one rule that prevents misreading everything

Original file = money GIVEN to candidates (contributions, small). Updated file = money SPENT on ads (independent expenditures, huge). If you ever describe the updated file's billions as 'donations' or 'contributions,' that's wrong, it's ad spending.

Part 3 — The dashboards, one by one

There are 8 dashboards. The first group uses the original contribution file; the second group uses the updated independent-expenditure file. Each section below explains what the dashboard is for, what each chart shows, and what to watch out for.

Group A — Contribution dashboards (original file)

1. `industry_rankings.html` — Corporate giving by industry

Purpose: **rank industries by how much their corporate PACs gave to candidates.** This is the dashboard your team likes.

It has three sections:

- Industry Share of Total Giving (donut): each slice is one industry's share of total matched corporate-PAC dollars. Quick way to see which industries dominate.
- Industry Ranking (horizontal bars): industries ordered biggest-to-smallest by dollars. Hovering shows how many firms and transactions are behind each bar.
- Top 3 Firms in Each Industry (grid of mini-charts): inside each industry, the leading firms. Many industries have only one firm, so a single bar is common.

What it means: **this is 'which industries' corporate PACs gave the most.** Top industries are Legal (\$139K, all Hall Render), Defense (\$110K), and Healthcare (\$101K). Caveat: many 'industries' are really one company, so read them as 'the top firm in this category' rather than a broad industry trend.

2. `industry_rankings_v2.html` — The refreshed version

Identical layout to #1, but rebuilt from the updated file using your `firms_master.csv` crosswalk, with one addition: a 'By type' toggle on the ranking bar chart that splits each industry into Direct contribution / Independent expenditure FOR / AGAINST.

What it means: the industries come out the same as the original (the contribution data didn't change). **The new toggle reveals that almost all corporate-PAC industry money is direct contributions — except Healthcare, whose money came as independent expenditures. A small but real finding: corporate PACs give; they rarely run ads.**

3. `monthly_by_firm.html` — Contributions over time

Purpose: show how corporate-PAC giving rises and falls month by month across the election cycle.

- Total Matched Giving by Month: the overall monthly heartbeat — low early, rising toward the election.
- Single Firm Drill-Down: pick one firm and see its monthly pattern.
- Top N Firms: compare the biggest givers over time on one chart.

- Industry Stack (Monthly): the same months, stacked by industry.
- All Firms (Mini Trends): a sparkline for every firm, with industry filter and search.

What it means: the expected election-cycle shape — giving ramps up before Election Day.

Note: this view is based on the small matched-contribution subset, so totals are modest.

4. election_comparison.html — Contributions across election dimensions

Purpose: compare contribution patterns across four lenses, each on its own tab: Primary vs General, Pre- vs Post-Election Day, Election Year vs Off-Year, and Office (House/Senate/President). Each can be sliced by industry, firm, or party. A final section, 'Recipients — Who Got the Money,' lists the specific candidates who received the contributions.

What it means: shows the timing and targeting of corporate contributions. Built on the contribution subset, so amounts are modest; the recipients section names candidates and the firms that funded them.

Group B — Independent-expenditure dashboards (updated file)

These five use the updated file. Remember: every figure here is Super PAC AD SPENDING (for or against candidates), not contributions.

5. ie_overview.html — The big picture of Super PAC ad spending

Purpose: the headline summary of the ~\$4.49B in independent expenditures.

- Summary cards: total spending, how much went to positive (FOR) vs attack (AGAINST) ads, and the 'attack-ad ratio.'
- FOR vs AGAINST by party: how much was spent supporting vs attacking each party's candidates.
- FOR vs AGAINST by office: President vs Senate vs House — and notably, the presidential race skews positive while Senate/House skew negative.
- Most-targeted candidates: top 25 candidates by total ad money for+against them (deduplicated across multiple FEC IDs). Harris tops it at \$1.1B.
- Top Super PACs: which committees spent the most (Future Forward \$510M, etc.).
- Contractor-vendors paid the most + by NAICS industry: which ad firms got the money.

What it means: this is the real story of 2024 money — Super PACs spending billions on ads, more than half of it negative.

6. ie_vendor_industries.html — Industries of the ad vendors

Purpose: the closest 'industry rankings' equivalent on the new file. But these are the industries of the VENDORS the Super PACs paid (classified by SAM.gov NAICS codes), NOT donor industries.

- Industries by spending: ranks vendor industries (Postal Service for direct mail, digital ad platforms, printing, PR, media buying). A toggle splits by FOR/AGAINST or by party.
- Industries by number of vendors: how many distinct firms make up each industry.
- Full detail table: every industry with totals, vendor counts, for/against, party split, and top firms.

What it means: **this is the political-advertising supply chain — who gets paid to make and run the ads**. It carries a correction note about the Meta mismatch (see Part 1's SAM.gov caution).

7. **ie_election.html — Ad spending across election dimensions**

The independent-expenditure counterpart to #4. Same four tabs (Primary vs General, Pre/Post Election Day, Year, Office), but now measuring ad SPENDING, sliced by party / office / FOR-vs-AGAINST, plus a deduplicated most-targeted-candidates chart.

What it means: **shows where and when the ad money was aimed**. The Primary-vs-General tab is dramatically lopsided toward the general election (\$3.76B vs \$691M) — Super PACs pour money into November, not primaries.

8. **ie_monthly.html — Ad spending over time, by Super PAC**

The independent-expenditure counterpart to #3. Monthly totals stacked by FOR vs AGAINST, plus the top Super PAC committees' spending over time (top 5/10/15 selector).

What it means: **the timing of the ad war**. Spending surges in the months before the November 2024 election; Future Forward is the largest single spender.

Part 4 — Old vs new: which dashboard to use

Because there are old/new pairs, here's how to choose:

- Want corporate-PAC GIVING by industry (the classic view your team likes)? Use `industry_rankings_v2.html`.
- Want the big picture of Super PAC AD spending? Use `ie_overview.html`.
- Want which ad VENDORS / vendor industries got paid? Use `ie_vendor_industries.html`.
- Want timing of ad spending? Use `ie_monthly.html` (ad spending) or `monthly_by_firm.html` (contributions).
- Want election-dimension comparisons? Use `ie_election.html` (ad spending) or `election_comparison.html` (contributions).

The mental shortcut

If the dollar figures are in the millions and about 'industries giving,' it's the contribution dashboards (Group A). If the figures are in the billions and about 'for/against ads,' it's the independent-expenditure dashboards (Group B).

Glossary — every term and code in one place

Organizations

- PAC (Political Action Committee): collects money and GIVES it to candidates, in small capped amounts.
- Super PAC: raises unlimited money and SPENDS it on independent ads; cannot give to candidates.
- Corporate PAC: a PAC sponsored by a company (funded by its employees).
- FEC: Federal Election Commission, the source of all this data.
- SAM.gov: the federal contractor registry used to flag vendors that are also government contractors.

Transaction-type codes

- 24K: direct contribution to a candidate (PAC gives money).
- 24E: independent expenditure FOR a candidate (Super PAC pays for a positive ad).
- 24A: independent expenditure AGAINST a candidate (Super PAC pays for an attack ad).
- 24C / 24Z: coordinated and in-kind/other expenditures (small amounts).

Election codes

- PGI: primary/general indicator. P = primary, G = general, C = convention, R = runoff, S = special.
- Office: House, Senate, or President.
- Party: DEM/DFL = Democrat, REP = Republican, IND = Independent.

Data terms

- CMTE_NM: the committee (PAC/Super PAC) that gave or spent the money.
- CAND_NAME / CAND_ID: the candidate the money targeted, and their unique FEC id.
- NAME: in independent-expenditure rows, this is the VENDOR the Super PAC paid (not a donor).
- NAICS: federal industry classification code, used to label vendor industries.
- firm_clean / industry: the donor company and its industry, from the firms_master.csv crosswalk.